

20261916 NH_300 .trees output

Protocol

1. Conda environment:

packages in environment at /home1/mahj/.conda/envs/tsdate_vcz:

#

# Name	Version	Build	Channel
_openmp_mutex	4.5	20_gnu	conda-forge
appdirs	1.4.4	pypi_0	pypi
asciitree	0.3.3	pypi_0	pypi
attrs	26.1.0	pyhcf101f3_0	conda-forge
bed-reader	1.1.0	pypi_0	pypi
bio2zarr	0.1.4	pypi_0	pypi
bzip2	1.0.8	hda65f42_10	conda-forge
ca-certificates	2026.7.22	hbd8a1cb_0	conda-forge
click	8.5.0	pypi_0	pypi
coloredlogs	15.0.1	pypi_0	pypi
cyvcf2	0.34.0	pypi_0	pypi
daiquiri	3.0.0	pyhd3deb0d_0	conda-forge
deprecated	1.3.1	pyhd8ed1ab_1	conda-forge
donfig	0.8.1.post1	pypi_0	pypi
fasteners	0.20	pyhd8ed1ab_0	conda-forge
google-crc32c	1.8.0	pypi_0	pypi
humanfriendly	10.0	pypi_0	pypi
humanize	4.16.0	pyhd8ed1ab_0	conda-forge
icu	78.3	py310h44b86e0_2	conda-forge
jsonschema	4.26.0	pyhcf101f3_1	conda-forge
jsonschema-specifications	2025.9.1	pyhcf101f3_0	conda-forge
ld_impl_linux-64	2.46.1	default_hbd61a6d_102	conda-forge
libblas	3.11.0	10_h4a7cf45_openblas	conda-forge
libcblas	3.11.0	10_h0358290_openblas	conda-forge
libexpat	2.8.1	hecca717_1	conda-forge
libffi	3.7.0	h81df57d_1	conda-forge
libgcc	16.2.0	ha9f2e26_4	conda-forge
libgfortran	16.2.0	h69a702a_4	conda-forge
libgfortran5	16.2.0	h6b99dfc_4	conda-forge
libgomp	16.2.0	he0feb66_4	conda-forge
liblapack	3.11.0	10_h47877c9_openblas	conda-forge
liblzma	5.8.3	hb03c661_1	conda-forge
libns1	2.0.1	hb9d3cd8_1	conda-forge
libopenblas	0.3.34	pthreads_hcf972fe_1	conda-forge
libsqlite	3.53.4	h13e7031_1	conda-forge
libstdcxx	16.2.0	h934c35e_4	conda-forge
libuuid	2.42.2	h5347b49_0	conda-forge
libxcrypt	4.4.38	h280c20c_0	conda-forge
libzlib	1.3.2	h25fd6f3_3	conda-forge
llvmlite	0.49.0	py311h1369f55_2	conda-forge
mpmath	1.4.1	pypi_0	pypi
msgpack-python	1.2.2	py311h4c6fd42_1	conda-forge

ncurses	6.6	hdb14827_1	conda-forge
numba	0.67.0	py311hdfc3925_1	conda-forge
numcodecs	0.15.1	py311hed34c8f_1	conda-forge
numpy	2.4.6	pypi_0	pypi
openssl	3.6.4	h781a0a9_0	conda-forge
packaging	26.3	pyhc364b38_0	conda-forge
pandas	3.0.5	pypi_0	pypi
pip	26.2.1	pyh8b19718_0	conda-forge
psutil	7.2.2	py311h194be0f_2	conda-forge
pysam	0.24.1	pypi_0	pypi
python	3.11.16	h8ab3286_0_cpython	conda-forge
python-dateutil	2.9.0.post0	pypi_0	pypi
python-json-logger	4.2.0	pyhd8ed1ab_0	conda-forge
python-lmdb	2.3.0	py311haee01d2_0	conda-forge
python_abi	3.11	8_cp311	conda-forge
pyyaml	6.0.3	pypi_0	pypi
readline	8.3	hd6e31c0_1	conda-forge
referencing	0.37.0	pyhcf101f3_0	conda-forge
rpds-py	2026.6.3	py311ha6e1976_1	conda-forge
scipy	1.16.3	py311hbe70eeb_2	conda-forge
setuptools	84.0.0	pyh332efcf_0	conda-forge
six	1.17.0	pypi_0	pypi
sortedcontainers	2.4.0	pyhd8ed1ab_1	conda-forge
tabulate	0.10.0	pypi_0	pypi
tk	8.6.13	noxft_h1df4ec4_4	conda-forge
tqdm	4.70.1	pypi_0	pypi
tsdate	0.2.7	pypi_0	pypi
tsinfer	0.5.1	py311h0372a8f_0	conda-forge
tskit	1.0.3	py311h0372a8f_0	conda-forge
typing_extensions	4.16.0	pyhcf101f3_0	conda-forge
tzdata	2026c	h151e31d_0	conda-forge
wheel	0.48.0	pyhd8ed1ab_0	conda-forge
wrapt	2.4.1	pypi_0	pypi
zarr	2.18.7	pyhd8ed1ab_0	conda-forge
zstd	1.5.7	hb78ec9c_7	conda-forge

2. Input data:

- Original files:
/project2/jitang_1167/projects/analyzeHawaiiPopGenome/NH-WGS/vcf_phased_kinship0.0884
- Sampled using: /project2/mahj_1946/variational_gamma/Scripts/sample_300_NH_WGS.sh
- Sampled data found in: /project2/mahj_1946/variational_gamma/Data/NH_300_WGS for chromosomes 2, 3, 5, 16
- All sample ids: /project2/mahj_1946/variational_gamma/Data/NH_300_WGS/all_samples.txt

- 300 randomly selected IDs (same across all chromosomes):

/project2/mahj_1946/variational_gamma/Data/NH_300_WGS/selected_300_samples.txt

- Ancestral state array:

/project2/chia657_28/resources/ancestral_alignments/human_ancestor_GRCh38_e114 , copied to

/project2/mahj_1946/variational_gamma/Data/homo_sapiens_ancestor_{2, 3, 5, 16}.fa

3. Intermediate workflow: ancestral state and sorting by allelic position

- Python script: /project2/mahj_1946/variational_gamma/Scripts/vcf_add_ancestral.py
- Bash script: /project2/mahj_1946/variational_gamma/Scripts/NH_300_WGS_ancestral.sh
- Intermediate output (sorted .vcf.gz with ancestral array):

/project2/mahj_1946/variational_gamma/Data/NH_300_WGS/NH_chr{2, 3, 5, 16}_300.ancestral.vcf.gz

4. Conversion from .vcf.gz and to .vcz

- Bash script: /project2/mahj_1946/variational_gamma/Scripts/NH_300_WGS_to_zarr.sh
- Output: /project2/mahj_1946/variational_gamma/Data/NH_300_WGS/NH_chr{2, 3, 5, 16}.final.vcz
- Zarr success:

```
==> NH_zarr_11998926_16.log <==
  Scan: 100%|██████████| 1.00/1.00 [00:00<00:00, 1.95files/s]
  Explode: 100%|██████████| 611k/611k [02:19<00:00, 4.39kvars/s]
  Encode: 100%|██████████| 1.24G/1.24G [02:50<00:00, 7.25MB/s]
  Finalise: 100%|██████████| 19.0/19.0 [00:28<00:00, 1.50s/array]

==> NH_zarr_11998926_2.log <==
  Scan: 100%|██████████| 1.00/1.00 [00:00<00:00, 2.92files/s]
  Explode: 100%|██████████| 1.97M/1.97M [07:15<00:00, 4.52kvars/s]
  Encode: 100%|██████████| 5.27G/5.27G [08:54<00:00, 9.86MB/s]
  Finalise: 100%|██████████| 19.0/19.0 [01:13<00:00, 3.88s/array]

==> NH_zarr_11998926_3.log <==
  Scan: 100%|██████████| 1.00/1.00 [00:00<00:00, 3.08files/s]
  Explode: 100%|██████████| 1.69M/1.69M [06:14<00:00, 4.51kvars/s]
  Encode: 100%|██████████| 3.82G/3.82G [07:43<00:00, 8.25MB/s]
  Finalise: 100%|██████████| 19.0/19.0 [01:10<00:00, 3.73s/array]

==> NH_zarr_11998926_5.log <==
  Scan: 100%|██████████| 1.00/1.00 [00:00<00:00, 2.54files/s]
  Explode: 100%|██████████| 1.50M/1.50M [05:43<00:00, 4.38kvars/s]
  Encode: 100%|██████████| 2.85G/2.85G [07:26<00:00, 6.39MB/s]
  Finalise: 100%|██████████| 19.0/19.0 [01:08<00:00, 3.60s/array]
```

5. Conversion from .vcz to .trees

- Python script: /project2/mahj_1946/variational_gamma/Scripts/vcz_to_trees.py
- Bash script: /project2/mahj_1946/variational_gamma/Scripts/NH_300_WGS_to_trees.sh
- Output: /project2/mahj_1946/variational_gamma/Analysis/NH_chr{2, 3, 5, 16}.trees
- .trees output:

```
(tsdate_vcz) bash-4.4$ du -sh *
122M    NH_chr16.trees
351M    NH_chr2.trees
300M    NH_chr3.trees
269M    NH_chr5.trees
(tsdate_vcz) bash-4.4$ pwd
/project2/mahj_1946/variational_gamma/Analysis
```

6. Log files:

- Detailed .log files can be found at:
/project2/mahj_1946/variational_gamma/Scripts/NH_300_trees_12080923_{2, 3, 5, 16}.log
- Something of note:

```
Input VCZ:      ../Data/NH_300_WGS/NH_chr16.final.vcz
Ancestral state: variant_AA
Dated tree seq: ../Analysis/NH_chr16.trees
Trees:          154880
Nodes:          253017
Edges:          1202906
Samples:         600
Sites:          611175
```

- Even for the smallest .vcf.gz and the smallest chromosome, there is an extremely high number of trees recorded in the sequence.